RTL to GDSii
Flow for a 8 bit alu

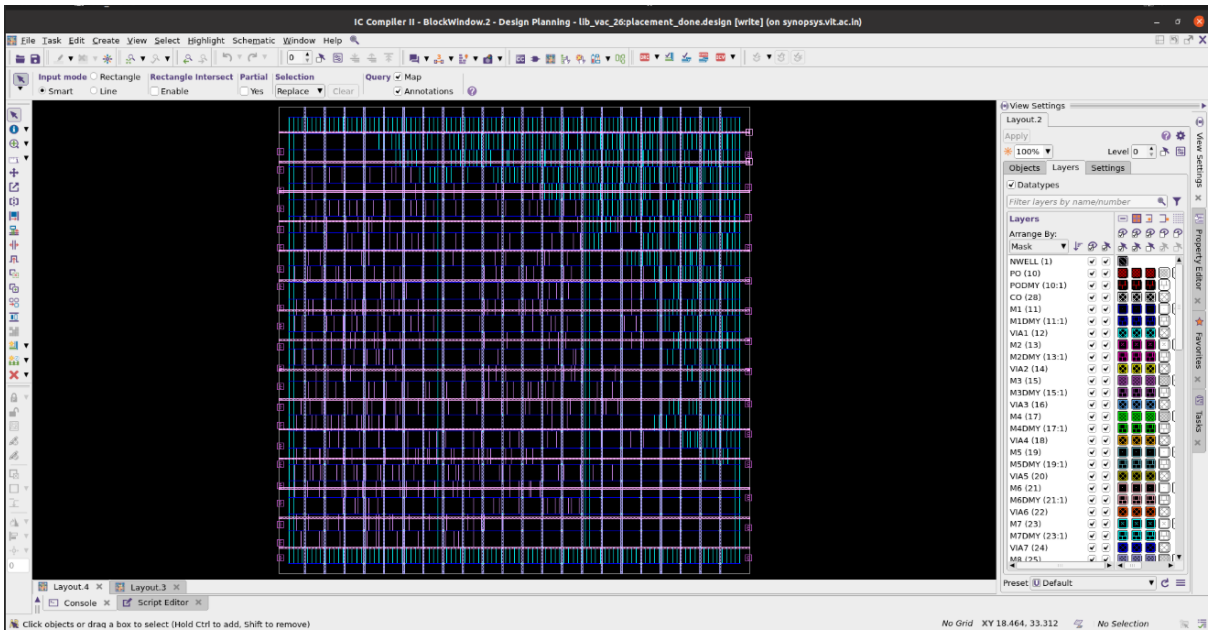
Report qor before

```
****************************************
Report : qor
         -summary
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 14:52:25 2026
****************************************
Information: The stitching and editing of coupling caps is turned OFF for design 'lib_vac_26:floorplan_done.design'. (TIM-125)
Information: Design Average RC for design floorplan_done  (NEX-011)
Information: r = 1.065882 ohm/um, via_r = 0.465866 ohm/cut, c = 0.071672 ff/um, cc = 0.000000 ff/um (X dir) (NEX-017)
Information: r = 1.392353 ohm/um, via_r = 0.580476 ohm/cut, c = 0.080723 ff/um, cc = 0.000000 ff/um (Y dir) (NEX-017)
Information: The RC mode used is VR for design 'alu'. (NEX-022)
Information: Update timing completed net estimation for all the timing graph nets (TIM-111)
Information: Net estimation statistics: timing graph nets = 286, routed nets = 0, across physical hierarchy nets = 0, parasitics cached nets = 286, delay annotated nets = 0, parasitics annotated nets = 0, multi-
voltage nets = 0. (TIM-112)

Timing
--------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------
func_slow          (Setup)            -7.82        -101.32              22
Design             (Setup)            -7.82        -101.32              22

func_fast          (Hold)             -1.08         -17.06              16
Design             (Hold)             -1.08         -17.06              16
--------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------
Cell Area (netlist):                                 635.87
Cell Area (netlist and physical only):              1061.31
Nets with DRC Violations:       147
1
```

Report qor after

```
****************************************
Report : qor
         -summary
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 14:55:37 2026
****************************************

Timing
--------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------
func_slow          (Setup)             0.00            0.00               0
Design             (Setup)             0.00            0.00               0

func_fast          (Hold)             -1.16          -19.68              30
Design             (Hold)             -1.16          -19.68              30
--------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------
Cell Area (netlist):                                 642.22
Cell Area (netlist and physical only):              1067.66
Nets with DRC Violations:       12
1
```

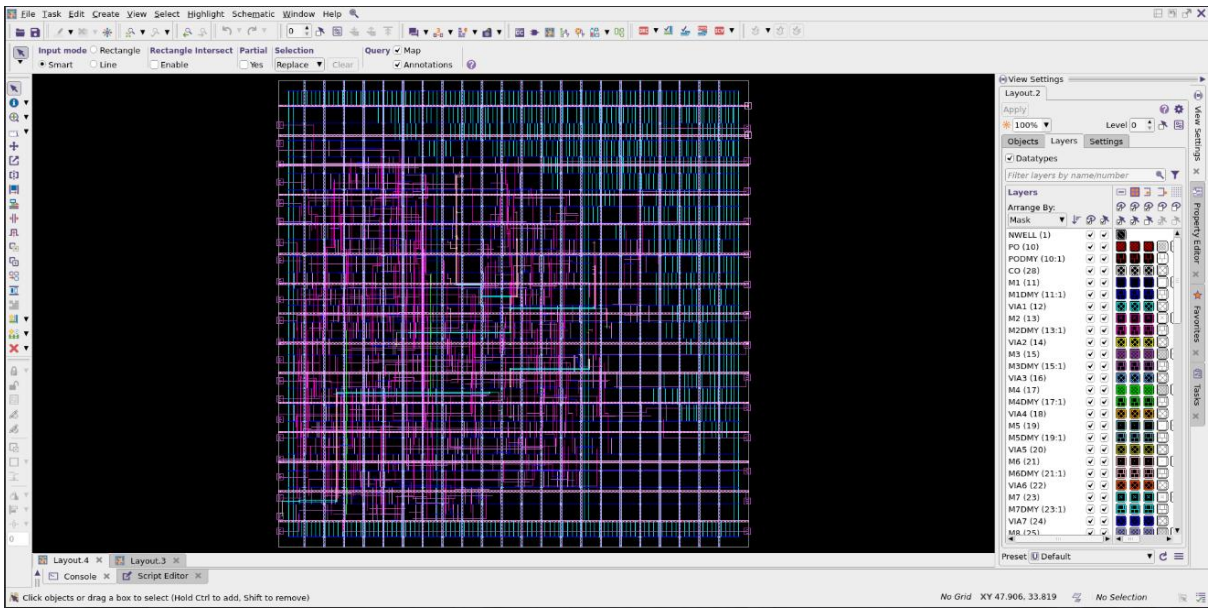
Placement so far



CTS report

```
Report : qor
         -summary
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:07:10 2026
****************************************

Timing
--------------------------------------------------------------------------
Context                                 WNS            TNS            NVE
--------------------------------------------------------------------------
func_slow          (Setup)            0.00           0.00              0
Design             (Setup)            0.00           0.00              0

func_fast          (Hold)             0.08           0.00              0
Design             (Hold)             0.08           0.00              0
--------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------
Cell Area (netlist):                                672.47
Cell Area (netlist and physical only):             1097.90
Nets with DRC Violations:        3
1
```

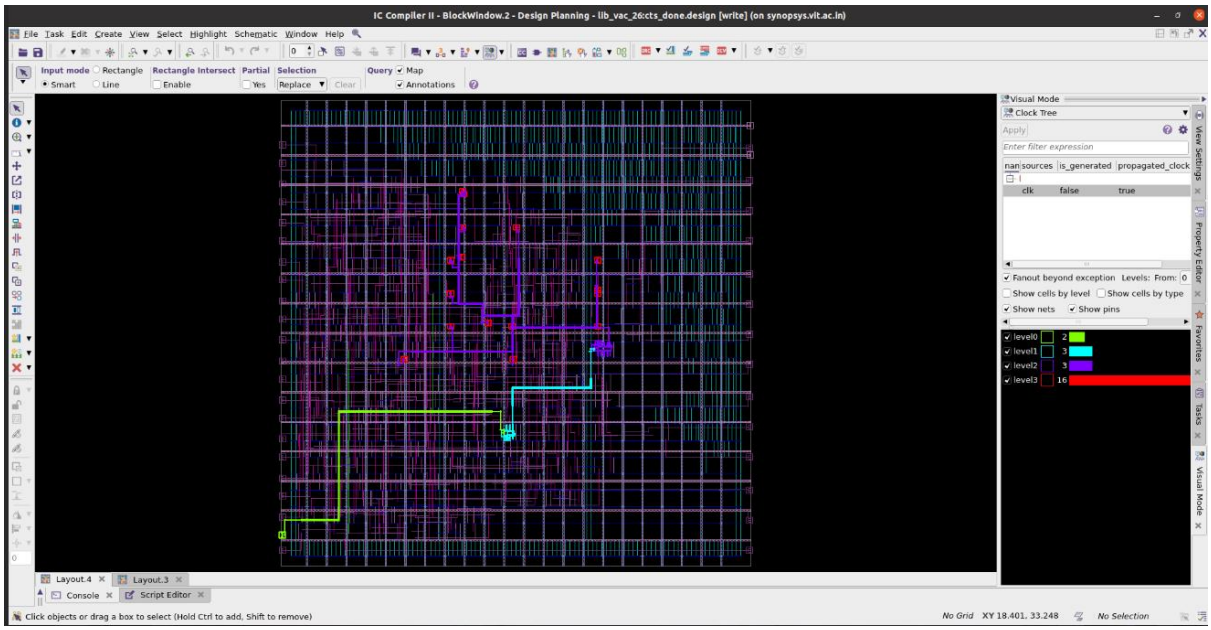
Placement so far

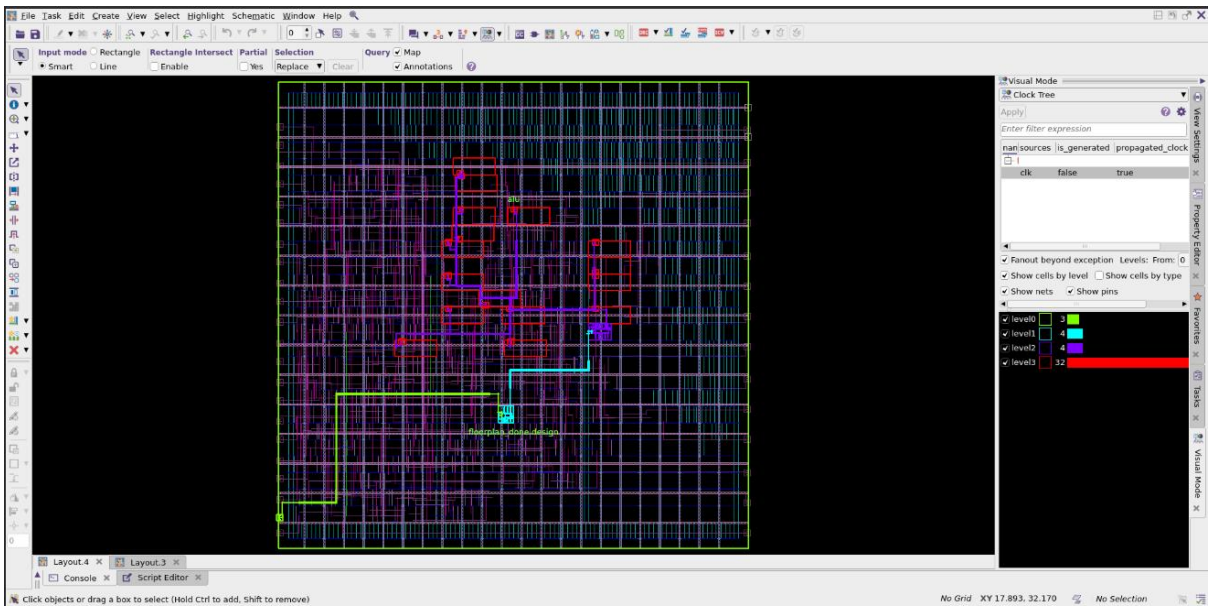


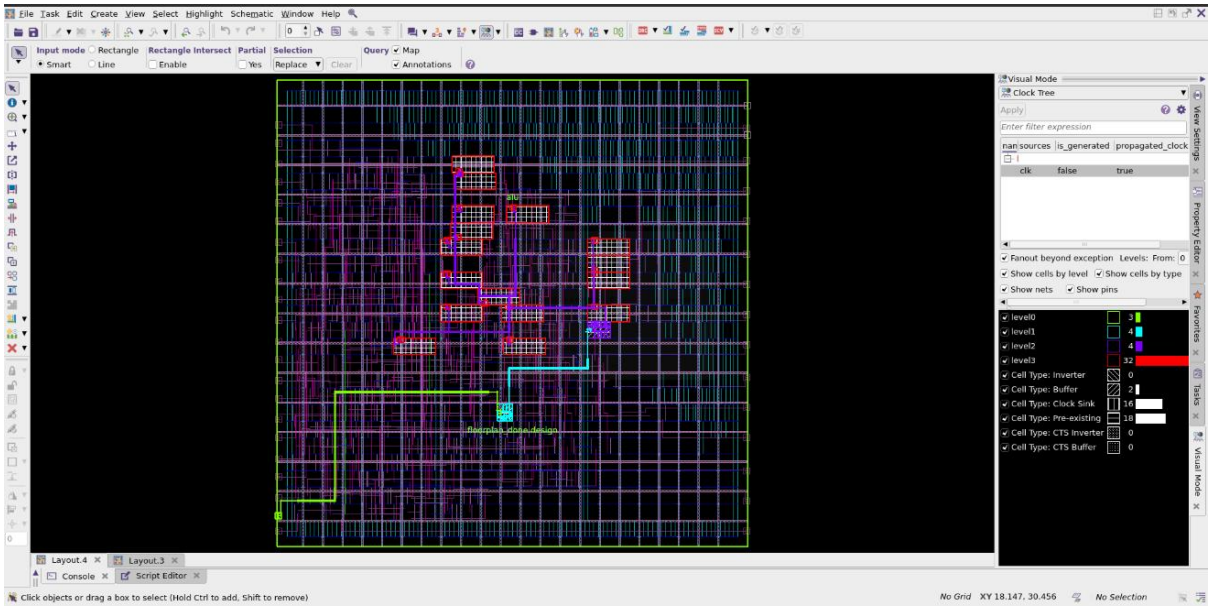
Clock Tree violations

```
****************************************
Report : constraint
         -all_violators
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:09:32 2026
****************************************

   late_timing
   -----------

Warning: Scenario func_fast is not configured for setup analysis: skipping. (UIC-058)
Error: None of the specified scenarios are active for setup analysis. (UIC-057)
Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
--------------------------------------------------------------------------------------------------
No paths.

   early_timing
   ------------

Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
--------------------------------------------------------------------------------------------------
No paths.

   Mode: func Corner: fast
   Scenario: func_fast
   max_transition
                              Required        Actual
   Net                       Transition     Transition        Slack  Violation
  ---------------------------------------------------------------------------
   aps_rename_5_                 0.40           0.41         -0.01  (VIOLATED)
     PIN : rb/out_reg_3_/Q       0.40           0.41         -0.01  (VIOLATED)
     PIN : copt_gre_mt_inst_2019/A
                                 0.40           0.41         -0.01  (VIOLATED)

   s_out[3]                      0.40           0.41         -0.01  (VIOLATED)
     PIN : sb/sh_reg_3_/Q        0.40           0.41         -0.01  (VIOLATED)
     PIN : mc1/copt_gre_mt_inst_2017/A
                                 0.40           0.41         -0.01  (VIOLATED)

   s_out[5]                      0.40           0.41         -0.01  (VIOLATED)
     PIN : sb/sh_reg_5_/Q        0.40           0.41         -0.01  (VIOLATED)
     PIN : mc1/copt_gre_mt_inst_2003/A
                                 0.40           0.41         -0.01  (VIOLATED)

  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 3
```

Violations


IC Compiler II - BlockWindow.2 - Design Planning - lib_vac_26:cts_done.design [write] (on synopsys.vit.ac.in)
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Rectangle Intersect Partial Selection Query Map
Smart Line Enable Yes Replace Clear Annotations
Visual Mode
Clock Tree
Apply
Enter filter expression
nan sources is_generated propagated_clock
clk false true
Fanout beyond exception Levels: From: 0
Show cells by level Show cells by type
Show nets Show pins
level0 2
level1 3
level2 3
level3 16
View Settings
Property Editor
Favorites
Tasks
Visual Mode
Layout.4
Layout.3
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 18.401, 33.248
No Selection


File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Rectangle Intersect Partial Selection Query Map
Smart Line Enable Yes Replace Clear Annotations
Visual Mode
Clock Tree
Apply
Enter filter expression
nan sources is_generated propagated_clock
clk false true
Fanout beyond exception Levels: From: 0
Show cells by level Show cells by type
Show nets Show pins
level0 3
level1 4
level2 4
level3 32
View Settings
Property Editor
Favorites
Tasks
Visual Mode
Layout.4
Layout.3
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 17.893, 32.170
No Selection



report clock tree violations

```
icc2_shell> report_clock_tree_options
****************************************
 Report : target skew and latency
 Design : alu
 Date   : Sun Mar 29 15:14:42 2026
****************************************

##Target Skew

Clock                        Corner                        Target
-----------------------------------------------------------------
clock (mode func)            fast                          0.6
clock (mode func)            slow                          0.6

##Target Latency

Clock                        Corner                        Target
-----------------------------------------------------------------
clock (mode func)            fast                          0.05

##Max Level Count

Clock                                   Count
-----------------------------------------------------------------
No max level constraints found.

##Root NDR Fanout Limit

Clock                                   Fanout limit
-----------------------------------------------------------------
No Root NDR fanout limit constraints applied.

##Trunk nets specific drivers

-----------------------------------------------------------------------------
1
```

report_clock_qor

```
Report : clock qor
        -type summary
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:16:05 2026
****************************************

Attributes
===========
M Master Clock
G Generated Clock
& Internal Generated Clock
U User Defined Skew Group
D Default Skew Group
* Generated Clock Balanced Separately

==========================================
==== Summary Reporting for Corner fast ====
==========================================

============================================================ Summary Table for Corner fast ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
-------------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func_fast
clock                                     M,D        16      3        0      0.00      6.35      0.16      0.00         0         0    140.01
-------------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                           16      3        0      0.00      6.35      0.16      0.00         0         0    140.01


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
==========================================
==== Summary Reporting for Corner slow ====
==========================================

============================================================ Summary Table for Corner slow ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
-------------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func_slow
clock                                     M,D        16      3        0      0.00      6.35      0.16      0.00         0         0    140.01
-------------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                           16      3        0      0.00      6.35      0.16      0.00         0         0    140.01


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
1
```

report_qor -summary

```
****************************************
Report : qor
        -summary
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:17:03 2026
****************************************

Timing
------------------------------------------------------------------------
Context                                 WNS            TNS            NVE
------------------------------------------------------------------------
func_slow          (Setup)            0.00           0.00              0
Design             (Setup)            0.00           0.00              0

func_fast          (Hold)             0.08           0.00              0
Design             (Hold)             0.08           0.00              0
------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------
Cell Area (netlist):                            672.47
Cell Area (netlist and physical only):         1097.90
Nets with DRC Violations:        3
1
```

report_timing -delay_type min

```
****************************************
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -report_by design
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:17:42 2026
****************************************

  Startpoint: rst (input port clocked by clock)
  Endpoint: sb/sh_reg_0_ (removal check against rising-edge clock clocked by clock)
  Mode: func
  Corner: fast
  Scenario: func_fast
  Path Group: INPUTS
  Path Type: min

  Point                                              Incr       Path
  -----------------------------------------------------------------------
  clock clock (rise edge)                            0.00       0.00
  clock network delay (propagated)                   1.10       1.12
  input external delay                               0.20       1.32 f
  rst (in)                                           0.02       1.32 f
  HFSINV_673_163/Y (INVX1_HVT)                       0.21       1.53 r
  copt_h_inst_2001/Y (NBUFFX4_HVT)                   0.28       1.81 r
  sb/sh_reg_0_/RSTB (DFFARX1_HVT)                    0.00       1.81 r
  data arrival time                                             1.81

  clock clock (rise edge)                            0.00       0.00
  clock network delay (propagated)                   0.36       0.36
  sb/sh_reg_0_/CLK (DFFARX1_HVT)                     0.00       0.36 r
  clock uncertainty                                  0.15       0.51
  library hold time                                  1.21       1.73
  data required time                                            1.73
  -----------------------------------------------------------------------
  data required time                                            1.73
  data arrival time                                            -1.81
  -----------------------------------------------------------------------
  slack (MET)                                                   0.08


1
```

report_timing -delay_type max

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:18:19 2026
****************************************

  Startpoint: sb/sh_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Endpoint: rb/out_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Mode: func
  Corner: slow
  Scenario: func_slow
  Path Group: REG2REG
  Path Type: max

  Point                                              Incr       Path
  -----------------------------------------------------------------------
  clock clock (rise edge)                            0.00       0.00
  clock network delay (propagated)                   0.36       0.36

  sb/sh_reg_7_/CLK (DFFARX1_HVT)                     0.00       0.36 r
  sb/sh_reg_7_/Q (DFFARX1_HVT)                       2.07       2.43 r
  mc1/copt_gre_mt_inst_2002/Y (NBUFFX2_LVT)          0.06       2.49 r
  mc1/ctmTdsLR_1_1061/Y (AOI22X1_LVT)                0.11       2.61 f
  mc2/ZINV_4_inst_605/Y (INVX0_HVT)                  0.07       2.68 r
  mc2/ctmTdsLR_1_1056/Y (AOI22X1_RVT)                0.20       2.89 f
  rb/HFSINV_4_81/Y (INVX0_LVT)                       0.03       2.92 r
  rb/out_reg_7_/D (DFFARX1_HVT)                      0.00       2.92 r
  data arrival time                                             2.92

  clock clock (rise edge)                            4.00       4.00
  clock network delay (propagated)                   0.26       4.26
  rb/out_reg_7_/CLK (DFFARX1_HVT)                    0.00       4.26 r
  clock uncertainty                                 -0.15       4.11
  library setup time                                -1.19       2.92
  data required time                                            2.92
  -----------------------------------------------------------------------
  data required time                                            2.92
  data arrival time                                            -2.92
  -----------------------------------------------------------------------
  slack (MET)                                                   0.00


1
```

report_timing -path_type full_clock_expanded

```
****************************************

  Startpoint: sb/sh_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Endpoint: rb/out_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Mode: func
  Corner: slow
  Scenario: func_slow
  Path Group: REG2REG
  Path Type: max

  Point                                              Incr       Path
  -----------------------------------------------------------------------
  clock clock (rise edge)                            0.00       0.00
  source latency                                     0.20       0.20
  clk (in)                                           0.00       0.20 r
  ctosc_dly_trglat_inst_1992/Y (NBUFFX4_RVT)         0.07       0.27 r
  ctosc_dly_trglat_inst_1991/Y (NBUFFX8_RVT)         0.10       0.36 r
  sb/sh_reg_7_/CLK (DFFARX1_HVT)                     0.00       0.36 r

  sb/sh_reg_7_/CLK (DFFARX1_HVT)                     0.00       0.36 r
  sb/sh_reg_7_/Q (DFFARX1_HVT)                       2.07       2.43 r
  mc1/copt_gre_mt_inst_2002/Y (NBUFFX2_LVT)          0.06       2.49 r
  mc1/ctmTdsLR_1_1061/Y (AOI22X1_LVT)                0.11       2.61 f
  mc2/ZINV_4_inst_685/Y (INVX0_HVT)                  0.07       2.68 r
  mc2/ctmTdsLR_1_1056/Y (AOI22X1_RVT)                0.20       2.89 f
  rb/HFSINV_4_81/Y (INVX0_LVT)                       0.03       2.92 r
  rb/out_reg_7_/D (DFFARX1_HVT)                      0.00       2.92 r
  data arrival time                                             2.92

  clock clock (rise edge)                            4.00       4.00
  source latency                                     0.10       4.10
  clk (in)                                           0.00       4.10 r
  ctosc_dly_trglat_inst_1992/Y (NBUFFX4_RVT)         0.07       4.17 r
  ctosc_dly_trglat_inst_1991/Y (NBUFFX8_RVT)         0.10       4.26 r
  rb/out_reg_7_/CLK (DFFARX1_HVT)                    0.00       4.26 r

  clock uncertainty                                 -0.15       4.11
  library setup time                                -1.19       2.92
  data required time                                            2.92
  -----------------------------------------------------------------------
  data required time                                            2.92
  data arrival time                                            -2.92
  -----------------------------------------------------------------------
  slack (MET)                                                   0.00


1
```

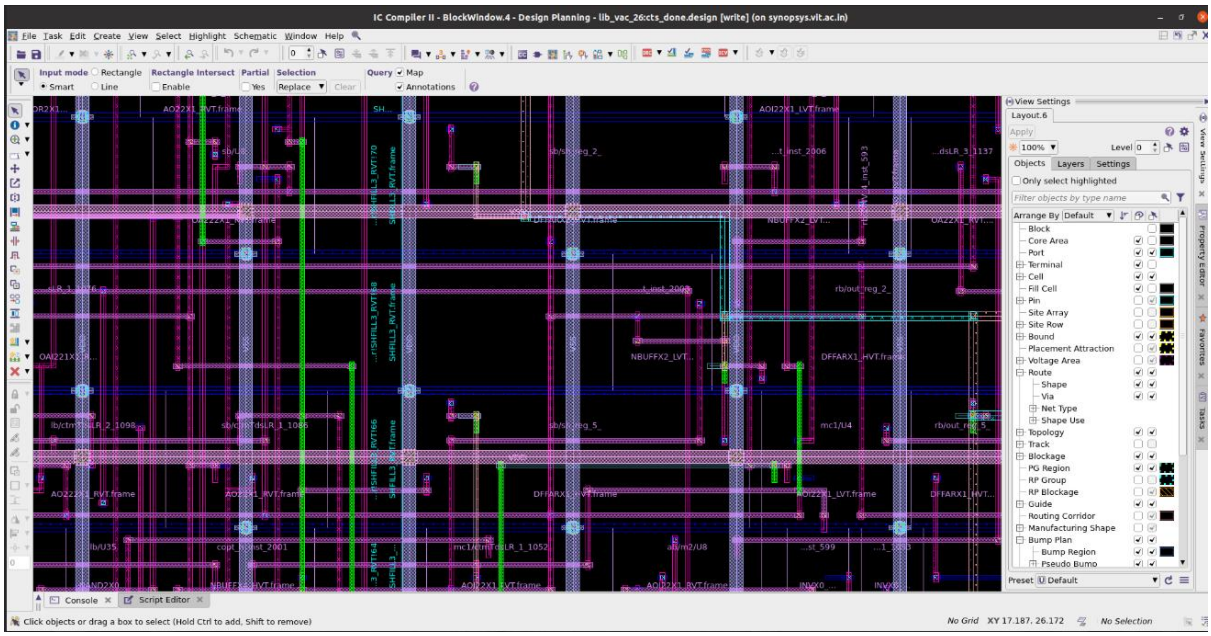
report_timing -delay_type min

```
****************************************
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -report_by design
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:23:28 2026
****************************************

  Startpoint: rst (input port clocked by clock)
  Endpoint: sb/sh_reg_0_ (removal check against rising-edge clock clocked by clock)
  Mode: func
  Corner: fast
  Scenario: func_fast
  Path Group: INPUTS
  Path Type: min

  Point                                              Incr       Path
  -----------------------------------------------------------------------
  clock clock (rise edge)                            0.00       0.00
  clock network delay (propagated)                   1.10       1.12
  input external delay                               0.20       1.32 f
  rst (in)                                           0.02       1.32 f
  HFSINV_673_163/Y (INVX1_HVT)                       0.21       1.53 r
  copt_h_inst_2001/Y (NBUFFX4_HVT)                   0.28       1.81 r
  sb/sh_reg_0_/RSTB (DFFARX1_HVT)                    0.00       1.81 r
  data arrival time                                             1.81

  clock clock (rise edge)                            0.00       0.00
  clock network delay (propagated)                   0.36       0.36
  sb/sh_reg_0_/CLK (DFFARX1_HVT)                     0.00       0.36 r
  clock uncertainty                                  0.15       0.51
  library hold time                                  1.21       1.73
  data required time                                            1.73
  -----------------------------------------------------------------------
  data required time                                            1.73
  data arrival time                                            -1.81
  -----------------------------------------------------------------------
  slack (MET)                                                   0.08


1
```

report_timing -delay_type max

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : alu
Version: V-2023.12-SP4
Date   : Sun Mar 29 15:24:07 2026
****************************************

  Startpoint: sb/sh_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Endpoint: rb/out_reg_7_ (rising edge-triggered flip-flop clocked by clock)
  Mode: func
  Corner: slow
  Scenario: func_slow
  Path Group: REG2REG
  Path Type: max

  Point                                            Incr       Path
  -----------------------------------------------------------------------
  clock clock (rise edge)                          0.00       0.00
  clock network delay (propagated)                 0.36       0.36

  sb/sh_reg_7_/CLK (DFFARX1_HVT)                   0.00       0.36 r
  sb/sh_reg_7_/Q (DFFARX1_HVT)                     2.07       2.43 r
  mc1/copt_gre_mt_inst_2002/Y (NBUFFX2_LVT)        0.06       2.49 r
  mc1/ctmTdsLR_1_1061/Y (AOI22X1_LVT)              0.11       2.61 f
  mc2/ZINV_4_inst_685/Y (INVX0_HVT)                0.07       2.68 r
  mc2/ctmTdsLR_1_1056/Y (AOI22X1_RVT)              0.20       2.89 f
  rb/HFSINV_4_81/Y (INVX0_LVT)                     0.03       2.92 r
  rb/out_reg_7_/D (DFFARX1_HVT)                    0.00       2.92 r
  data arrival time                                           2.92

  clock clock (rise edge)                          4.00       4.00
  clock network delay (propagated)                 0.26       4.26
  rb/out_reg_7_/CLK (DFFARX1_HVT)                  0.00       4.26 r
  clock uncertainty                               -0.15       4.11
  library setup time                              -1.19       2.92
  data required time                                          2.92
  -----------------------------------------------------------------------
  data required time                                          2.92
  data arrival time                                          -2.92
  -----------------------------------------------------------------------
  slack (MET)                                                 0.00


1
```

routing

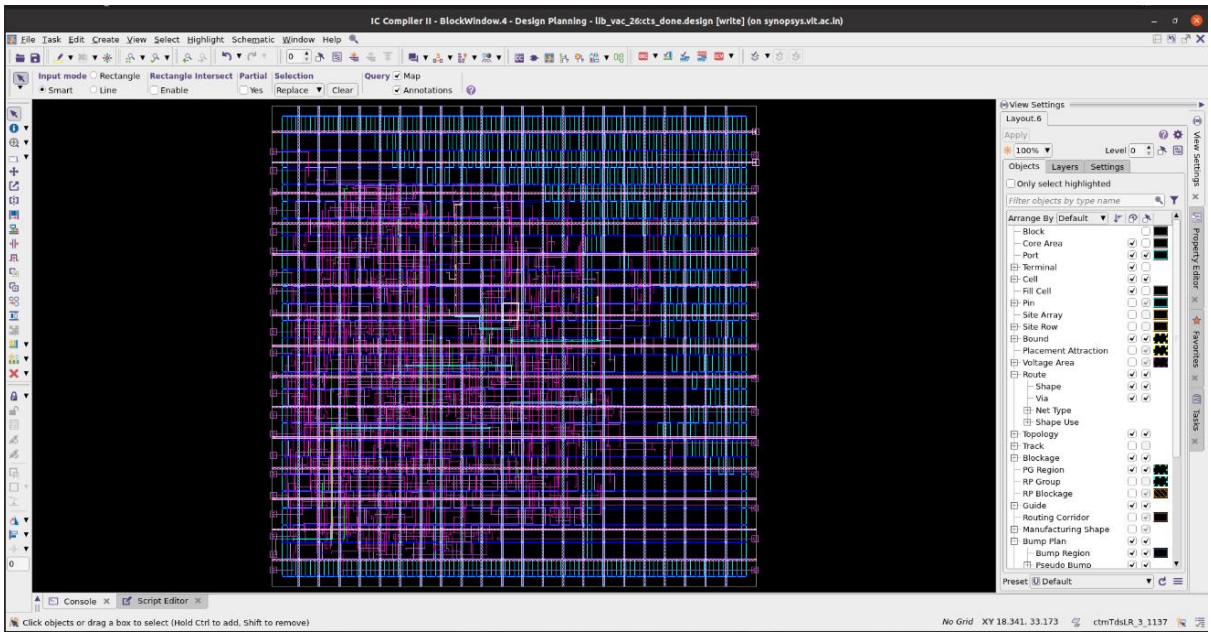


report_constrainsts -all_violators optimization

```
  lb/n22                          0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U37/Y                0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U38/A2               0.40          0.47          -0.07  (VIOLATED)

  lb/n10                          0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U25/Y                0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U26/A2               0.40          0.47          -0.07  (VIOLATED)

  lb/n50                          0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U60/Y                0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U61/A2               0.40          0.47          -0.07  (VIOLATED)

  lb/n35                          0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U48/Y                0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U49/A2               0.40          0.47          -0.07  (VIOLATED)

  lb/n41                          0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U54/Y                0.40          0.47          -0.07  (VIOLATED)
    PIN : lb/U55/A2               0.40          0.47          -0.07  (VIOLATED)

  ab/sum[0]                       0.40          0.46          -0.06  (VIOLATED)
    PIN : ab/h1/U1/Y              0.40          0.46          -0.06  (VIOLATED)
    PIN : ab/U37/A4               0.40          0.46          -0.06  (VIOLATED)
    PIN : ab/m2/U6/A4             0.40          0.46          -0.06  (VIOLATED)

  ab/sum1[7]                      0.40          0.42          -0.02  (VIOLATED)
    PIN : ab/U41/SO               0.40          0.42          -0.02  (VIOLATED)
    PIN : ab/m2/U3/A              0.40          0.42          -0.02  (VIOLATED)
    PIN : ab/f7/U1/A              0.40          0.42          -0.02  (VIOLATED)

 ---------------------------------------------------------------------------
  Number of max_transition violation(s): 114



  Mode: func Corner: fast
  Scenario: func_fast
 ---------------------------------------------------------------------------
  Number of min_capacitance violation(s): 0

 Mode: func
 Corner: fast
 Scenario: func_fast

 ---------------------------------------------------------------------------
  Number of min_pulse_width violation(s): 0
  Total number of violation(s): 114
```

```
Information: Timer using 'PrimeTime Delay Calculation, SI, Timing Window Analysis'. (TIM-050)
Endpoint                          Path Delay     Path Required       CRP    Slack Group    Scenario
-----------------------------------------------------------------------------------------------------
No paths.

   Mode: func Corner: fast
   Scenario: func_fast
   max_transition
                                 Required        Actual
   Net                          Transition     Transition        Slack  Violation
  ---------------------------------------------------------------------------
   s_out[5]                        0.40           0.41          -0.01  (VIOLATED)
     PIN : sb/sh_reg_5_/Q          0.40           0.41          -0.01  (VIOLATED)
     PIN : mc1/ropt_mt_inst_2022/A
                                   0.40           0.41          -0.01  (VIOLATED)

   HFSNET_18                       0.40           0.41          -0.01  (VIOLATED)
     PIN : HFSINV_673_163/Y        0.40           0.41          -0.01  (VIOLATED)
     PIN : copt_h_inst_2001/A
                                   0.40           0.41          -0.01  (VIOLATED)

   s_out[0]                        0.40           0.40          -0.00  (VIOLATED)
     PIN : sb/sh_reg_0_/Q          0.40           0.40          -0.00  (VIOLATED)
     PIN : mc1/ropt_mt_inst_2023/A
                                   0.40           0.40          -0.00  (VIOLATED)

   aps_rename_6_                   0.40           0.40          -0.00  (VIOLATED)
     PIN : rb/out_reg_2_/Q         0.40           0.40          -0.00  (VIOLATED)
     PIN : ropt_mt_inst_2025/A
                                   0.40           0.40          -0.00  (VIOLATED)

  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 4



   Mode: func Corner: fast
   Scenario: func_fast
  ---------------------------------------------------------------------------
   Number of min_capacitance violation(s): 0

  Mode: func
  Corner: fast
  Scenario: func_fast

  ---------------------------------------------------------------------------
   Number of min_pulse_width violation(s): 0
   Total number of violation(s): 4
1
```

Placement so far



paracitics extracted

```
*SPEF "IEEE 1481-1999"
*DESIGN "route_done"
*DATE "Sat Apr  4 10:05:57 2026"
*VENDOR "Synopsys Inc."
*PROGRAM "StarRC"
*VERSION "U-2022.12-SP5-2"
*DESIGN_FLOW "PIN_CAP NONE" "NAME_SCOPE LOCAL"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1.0 NS
*C_UNIT 1.0 FF
*R_UNIT 1.0 OHM
*L_UNIT 1.0 HENRY

// COMMENTS

// CORNER_NAME
// OPERATING_TEMPERATURE 25
// GLOBAL_TEMPERATURE 25
// TCAD_GRD_FILE ../tech/saed32nm_1p9m_Cmax.lv.nxtgrd
//   TCAD_TIME_STAMP Wed Jul 24 15:44:14 2019
//   TCADGRD_VERSION 84
// galaxy_comments_begin
//   local_nodeid
// galaxy_comments_end

*NAME_MAP
*289 f[7]
*290 f[6]
*291 f[5]
*292 f[4]
*293 f[3]
*294 f[2]
*295 f[1]
*296 f[0]
*297 z
*298 c
*299 v
*300 agtb
*301 altb
*302 aeqb
*303 a[7]
*304 a[6]
*305 a[5]
*306 a[4]
*307 a[3]
*308 a[2]
```

report _ parasitic

```
pt_shell> report_qor -summary
****************************************
Report : qor
        -summary
Design : alu
Version: V-2023.12-SP4
Date   : Sat Apr  4 10:30:16 2026
****************************************

  Summary of timing violations
  --------------------------------------------------------------------------------
  No setup violations found
  Hold  WNS:   -0.79  TNS:     -6.14  Number of Violating Paths:      8
  --------------------------------------------------------------------------------

  Area
  ------------------------------------------
  Net Interconnect area:                126.02
  Total cell area:                      662.05
  Design Area:                          788.06
  ------------------------------------------


  Cell & Pin Count
  ------------------------------------------
  Pin Count:                               901
  Hierarchical Cell Count:                  17
  Hierarchical Port Count:                 263
  Leaf Cell Count:                         256
  ------------------------------------------


  Design Rule Violations
  ------------------------------------------
  Total No. of Pins in Design:             901
  Total DRC Cost:                         0.00
  ------------------------------------------

1
```

reportannoted_parasitics

```
(lower bound)
----------------------------------------
0.076214            214       75.09%
0.990985            38        13.33%
1.905756            7         2.46%
2.820527            5         1.75%
3.735298            4         1.40%
4.650069            10        3.51%
5.564840            4         1.40%
6.479611            0         0.00%
7.394382            2         0.70%
8.309152            0         0.00%
9.223923            1         0.35%


Total R*C           Networks  Percentage
(lower bound)
----------------------------------------
0.000000            1         0.35%
0.000000            259       90.88%
0.000185            5         1.75%
0.000370            8         2.81%
0.000556            8         2.81%
0.000741            2         0.70%
0.000926            0         0.00%
0.001111            0         0.00%
0.001296            0         0.00%
0.001481            1         0.35%
0.001667            0         0.00%
0.001852            1         0.35%


Caps per pin        Networks  Percentage
(lower bound)
----------------------------------------
1.000000            16        5.61%
2.250000            65        22.81%
3.500000            151       52.98%
4.750000            36        12.63%
6.000000            12        4.21%
7.250000            1         0.35%
8.500000            1         0.35%
11.000000           1         0.35%
12.250000           1         0.35%
13.500000           1         0.35%


1
icc2_shell>
```

report_analysis_coverage

```
pt_shell> report_annotated_parasitics
****************************************
Report : annotated_parasitics
        -check
        -internal_nets
        -boundary_nets
        -driverless_nets
        -loadless_nets
        -pin_to_pin_nets
Design : alu
Version: V-2023.12-SP4
Date   : Sat Apr  4 10:34:03 2026
****************************************

                    |         |         |         |   RC    |   Not   |
Net Type            |  Total  | Lumped  |  RC pi  | network |Annotated|
--------------------+---------+---------+---------+---------+---------+
Internal nets       |         |         |         |         |         |
  - Pin to pin nets |     249 |       0 |       0 |     249 |       0 |
  - Driverless nets |       8 |       0 |       0 |       0 |       8 |
  - Loadless nets   |       0 |       0 |       0 |       0 |       0 |
--------------------+---------+---------+---------+---------+---------+
Boundary/port nets  |         |         |         |         |         |
  - Pin to pin nets |      36 |       0 |       0 |      36 |       0 |
  - Driverless nets |       0 |       0 |       0 |       0 |       0 |
  - Loadless nets   |       0 |       0 |       0 |       0 |       0 |
--------------------+---------+---------+---------+---------+---------+
                    |     293 |       0 |       0 |     285 |       8 |

1
pt_shell>
```

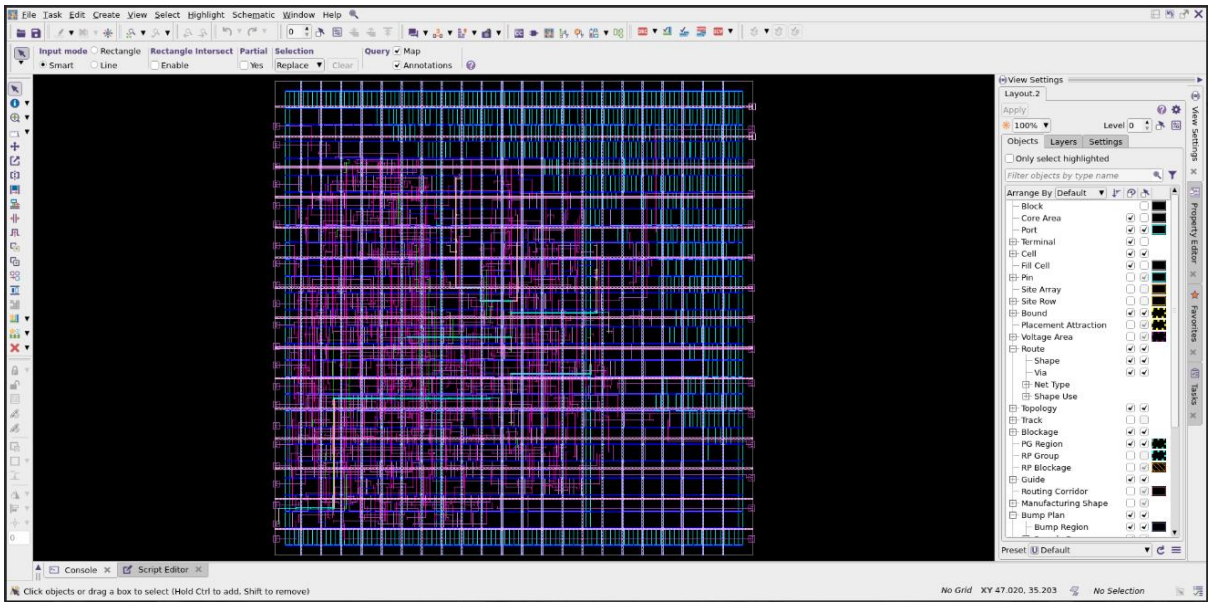
fixed

```
pt_shell> report_analysis_coverage
****************************************
Report : analysis_coverage
Design : alu
Version: V-2023.12-SP4
Date   : Sat Apr  4 10:42:03 2026
****************************************

Type of Check         Total              Met         Violated         Untested
-----------------------------------------------------------------------------------
setup                    16        16 (100%)         0 (  0%)         0 (  0%)
hold                     16        16 (100%)         0 (  0%)         0 (  0%)
recovery                 16        16 (100%)         0 (  0%)         0 (  0%)
removal                  16        16 (100%)         0 (  0%)         0 (  0%)
min_pulse_width          48        32 ( 67%)         0 (  0%)        16 ( 33%)
out_setup                14        14 (100%)         0 (  0%)         0 (  0%)
out_hold                 14        14 (100%)         0 (  0%)         0 (  0%)
-----------------------------------------------------------------------------------
All Checks              140       124 ( 89%)         0 (  0%)        16 ( 11%)

1
```

timing_eco.tcl ---- > buffers introdudced

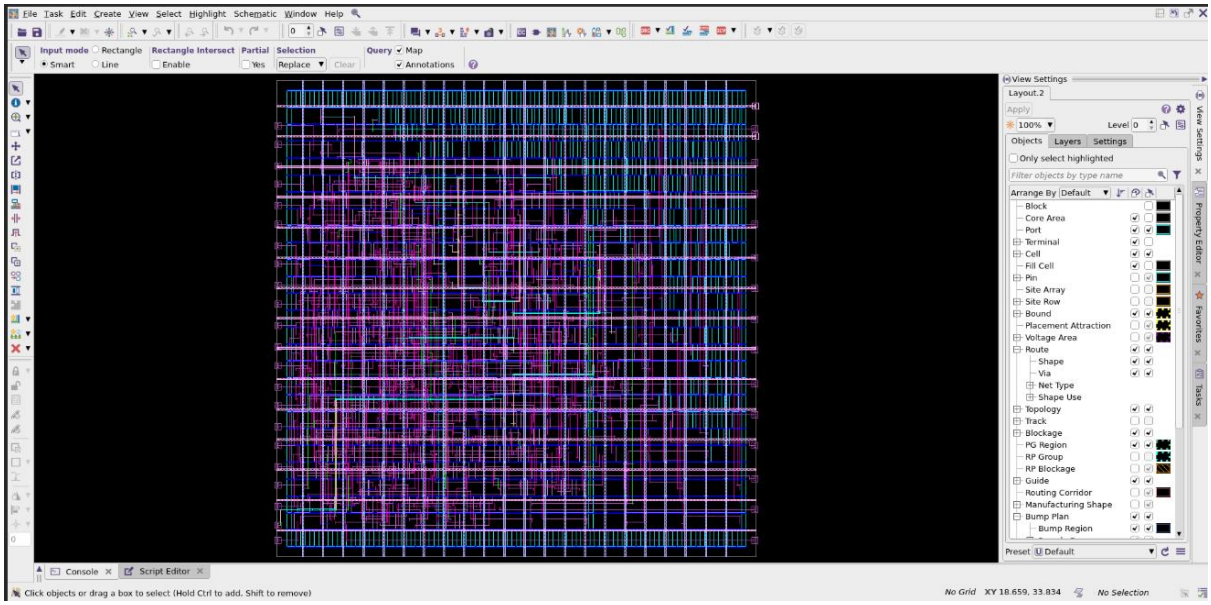
```
##############################################################################
# Change list, formatted for IC Compiler
#
#
#
##############################################################################
current_instance
insert_buffer [get_ports {f[4]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET1} -new_cell_names {U_PTECO_HOLD_BUF1}
insert_buffer [get_ports {f[3]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET2} -new_cell_names {U_PTECO_HOLD_BUF2}
insert_buffer [get_ports {f[7]}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET3} -new_cell_names {U_PTECO_HOLD_BUF3}
insert_buffer [get_ports {f[0]}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET4} -new_cell_names {U_PTECO_HOLD_BUF4}
insert_buffer [get_ports {f[1]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET5} -new_cell_names {U_PTECO_HOLD_BUF5}
insert_buffer [get_ports {f[5]}] DELLN2X2_HVT -new_net_names {net_PTECO_HOLD_NET6} -new_cell_names {U_PTECO_HOLD_BUF6}
insert_buffer [get_ports {f[6]}] DELLN2X2_HVT -new_net_names {net_PTECO_HOLD_NET7} -new_cell_names {U_PTECO_HOLD_BUF7}
insert_buffer [get_ports {f[2]}] DELLN2X2_HVT -new_net_names {net_PTECO_HOLD_NET8} -new_cell_names {U_PTECO_HOLD_BUF8}
size_cell {copt_gre_mt_inst_2010} {DELLN3X2_HVT}
size_cell {copt_gre_mt_inst_2008} {DELLN3X2_HVT}
size_cell {copt_gre_mt_inst_2021} {DELLN1X2_HVT}
size_cell {copt_gre_mt_inst_2009} {DELLN1X2_HVT}
size_cell {ropt_mt_inst_2025} {DELLN1X2_HVT}
insert_buffer [get_ports {f[7]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET9} -new_cell_names {U_PTECO_HOLD_BUF9}
insert_buffer [get_ports {f[0]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET10} -new_cell_names {U_PTECO_HOLD_BUF10}
insert_buffer [get_ports {f[1]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET11} -new_cell_names {U_PTECO_HOLD_BUF11}
insert_buffer [get_ports {f[6]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET12} -new_cell_names {U_PTECO_HOLD_BUF12}
insert_buffer [get_ports {f[2]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET13} -new_cell_names {U_PTECO_HOLD_BUF13}
insert_buffer [get_ports {f[3]}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET14} -new_cell_names {U_PTECO_HOLD_BUF14}
insert_buffer [get_ports {f[5]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET15} -new_cell_names {U_PTECO_HOLD_BUF15}
insert_buffer [get_ports {f[4]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET16} -new_cell_names {U_PTECO_HOLD_BUF16}
insert_buffer [get_ports {f[1]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET17} -new_cell_names {U_PTECO_HOLD_BUF17}
insert_buffer [get_ports {f[3]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET18} -new_cell_names {U_PTECO_HOLD_BUF18}
insert_buffer [get_ports {f[4]}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET19} -new_cell_names {U_PTECO_HOLD_BUF19}
insert_buffer [get_pins {U_PTECO_HOLD_BUF6/A}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET20} -new_cell_names {U_PTECO_HOLD_BUF20}
insert_buffer [get_pins {U_PTECO_HOLD_BUF7/A}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET21} -new_cell_names {U_PTECO_HOLD_BUF21}
insert_buffer [get_pins {U_PTECO_HOLD_BUF8/A}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET22} -new_cell_names {U_PTECO_HOLD_BUF22}
insert_buffer [get_pins {U_PTECO_HOLD_BUF8/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET23} -new_cell_names {U_PTECO_HOLD_BUF23}
insert_buffer [get_pins {HFSBUF_4_98/Y}] DELLN1X2_LVT -new_net_names {net_PTECO_HOLD_NET24} -new_cell_names {U_PTECO_HOLD_BUF24}
insert_buffer [get_pins {U_PTECO_HOLD_BUF7/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET25} -new_cell_names {U_PTECO_HOLD_BUF25}
insert_buffer [get_pins {U_PTECO_HOLD_BUF6/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET26} -new_cell_names {U_PTECO_HOLD_BUF26}
insert_buffer [get_pins {ropt_mt_inst_2024/Y}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET27} -new_cell_names {U_PTECO_HOLD_BUF27}
insert_buffer [get_pins {ropt_mt_inst_2026/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET28} -new_cell_names {U_PTECO_HOLD_BUF28}
insert_buffer [get_pins {U_PTECO_HOLD_BUF3/A}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET29} -new_cell_names {U_PTECO_HOLD_BUF29}
insert_buffer [get_pins {U_PTECO_HOLD_BUF4/A}] DELLN3X2_HVT -new_net_names {net_PTECO_HOLD_NET30} -new_cell_names {U_PTECO_HOLD_BUF30}
insert_buffer [get_pins {U_PTECO_HOLD_BUF4/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET31} -new_cell_names {U_PTECO_HOLD_BUF31}
insert_buffer [get_pins {U_PTECO_HOLD_BUF3/Y}] NBUFFX16_HVT -new_net_names {net_PTECO_HOLD_NET32} -new_cell_names {U_PTECO_HOLD_BUF32}
```

placed all the changes after routing

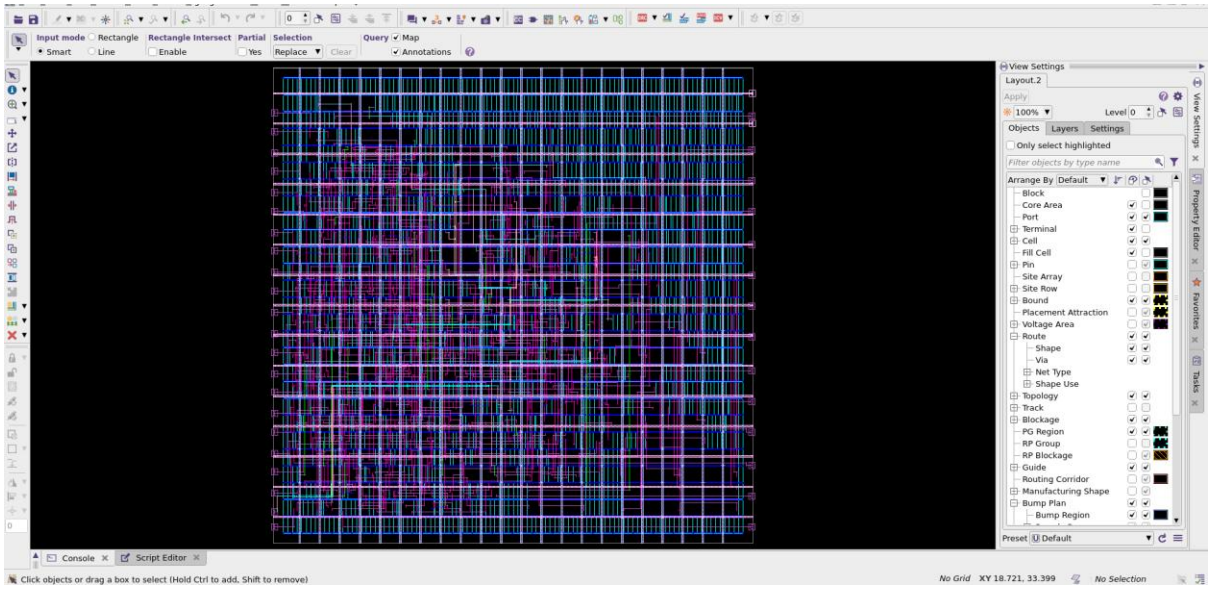


legality succeeded

```
        0          0          0    A cell violates a pnet.

        0          0          0    A cell is illegal at a site.
          0          0          0    A cell violates pin-track alignment rules.
          0          0          0    A cell is illegal at a site.
          0          0          0    A cell violates legal index rule.
          0          0          0    A cell has the wrong variant for its location.

        0          0          0    A cell is not aligned with a site.
          0          0          0    A cell is not aligned with the base site.
          0          0          0    A cell is not aligned with an overlaid site.

        0          0          0    A cell has an illegal orientation.

        0          0          0    A cell spacing rule is violated.
          0          0          0    A spacing rule is violated in a row.
          0          0          0    A spacing rule is violated between adjacent rows.
          0          0          0    A cell violates vertical abutment rule.
          0          0          0    A cell violates metal spacing rule.

        0          0          0    A layer rule is violated.
          0          0          0    A layer VTH rule is violated.
          0          0          0    A layer OD rule is violated.
          0          0          0    A layer OD max-width rule is violated.
          0          0          0    A layer ALL_OD corner rule is violated.
          0          0          0    A layer max-vertical-length rule is violated.
          0          0          0    A layer TPO rule is violated.
          0          0          0    Filler cell insertion cannot satisfy layer rules.

        0          0          0    A cell is in the wrong region.
          0          0          0    A cell is outside its hard bound.
          0          0          0    A cell is in the wrong voltage area.
          0          0          0    A cell violates an exclusive movebound.

        0          0          0    Two cells violate cts margins.

        0          0          0    Two cells violate coloring.


check_legality for block design alu succeeded!


check_legality succeeded.

**************************

1
```



Generating GDSii file

```
icc2_shell> report_utilization -config core_utilization
****************************************
Report : report_utilization
Design : alu
Version: V-2023.12-SP4
Date   : Sat Apr  4 12:26:34 2026
****************************************
Utilization Ratio:                      1.0000
Utilization options:
 - Area calculation based on:           core_area of block route_done
 - Categories of objects excluded:      None
Total Area:                             2058.5664
Total Capacity Area:                    2058.5664
Total Area of cells:                    2058.5664

1.0000
icc2_shell> save_block -as final
Information: The command 'save_block' cleared the undo history. (UNDO-016)
Information: Saving 'lib_vac_26:route_done.design' to 'lib_vac_26:final.design'. (DES-028)
1
icc2_shell> write_gds alu_final.gds
1
icc2_shell>
```

FINAL CHIP LAYOUT